

Trainer Debrief — Lab 21

Analyze Tollgate — Proving Root Cause and Telling the Data Story

Case study: Meridian Medical Devices — seal weld burst pressure

Objective: Consolidate your analysis into a defensible root cause conclusion (A3, K1).

***TRAINER ONLY.** Do not hand this out before the lab is complete.*

Expected analysis

lab-21-analyze-tollgate-evidence.xlsx

Proven: clamp pressure ($p < 0.001$), dwell time ($p < 0.001$), electrode tip age ($p < 0.001$), supplier C (Tukey, $p < 0.01$), the Suzhou-shift-C interaction ($p < 0.001$) and line-to-line as the dominant multi-vari family. NOT proven: ambient temperature ($p \sim 0.09$) and humidity (not significant, and collinear with temperature). Disproving causes matters as much as proving them — it is what stops the team spending SGD 200k on shop-floor air conditioning. Note the distinction the sponsor will probe: a 0.11 kPa/degC temperature effect could become statistically significant with a big enough sample and still be practically worthless against an 80 kPa spec width.

Every dataset is generated from a fixed seed, so these figures are reproducible — re-run the analysis on the shipped workbook and you will get these numbers.

What good looks like

Every candidate cause carries a verdict, proven causes are triangulated across multiple methods and quantified in dollars, and you have rehearsed a prepared response to each expected challenge.

Where this sits in the capstone

The Analyze tollgate. Consolidate everything from Labs 15-20 into a proven root cause set, translate the statistics into business language, and rehearse the challenges you will face. Sponsors do not fund p-values; they fund decisions. CAPSTONE BUILDING BLOCK — the output of this lab goes directly into your capstone project pack, which you present to the steering committee on Day 5.

The running process — figures that must reconcile

Checkpoint	Value	Established in
Baseline defect rate	2.9%	Lab 3
Measurement system	FAILS Gage R&R (~40% study variation)	Lab 11
Baseline capability	Pp 0.62 / Ppk 0.53	Lab 14
Proven Xs	clamp pressure, dwell time, electrode tip age	Labs 18-21
DOE optimum	clamp 47.4 bar, dwell 2.76 s	Labs 23-25
Improved capability	Ppk 1.40	Lab 27

Annual hard benefit	SGD 487,000 vs SGD 812,000 COPQ	Labs 1, 30
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If a learner's number disagrees with the column above, they have either used their own workplace data (fine — say so) or made an error (find it).

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